

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P2HR Worked Solutions

Jan 2020 | P2HR

33 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jan 2020 P2HR - Jan 2020 | P2HR

- 1** (a) Write $5^{17} \times 5^2$ as a single power of 5

.....
(1)

- (b) Write 800 as a product of its prime factors.
Show your working clearly.

.....
(2)

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1** **Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use the index law**Use the index law $a^m \times a^n = a^{m+n}$:

$$5^{17} \times 5^2 = 5^{19}$$

2. Find the LCM

Now write 800 as a product of prime factors:

$$800 = 8 \times 100$$

$$800 = 2^3 \times 2^2 \times 5^2$$

$$800 = 2^5 \times 5^2$$

FINAL ANSWER(a) 5^{19} , (b) $2^5 \times 5^2$

PAST PAPER QUESTION**Q#: 2** **Marks: 3****TOPIC: Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

- 2 The table gives information about the amount of money, in £, that Fiona spent in a grocery store each week during 2019

Amount spent (£x)	Frequency
$0 \leq x < 20$	5
$20 \leq x < 40$	11
$40 \leq x < 60$	8
$60 \leq x < 80$	19
$80 \leq x < 100$	9

Work out an estimate for the total amount of money that Fiona spent in the grocery store during 2019

£.....

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use midpoints

Use midpoints 10, 30, 50, 70, 90.

$$10(5) + 30(11) + 50(8) + 70(19) + 90(9)$$

$$= 50 + 330 + 400 + 1330 + 810 = 2920$$

FINAL ANSWER

£2920.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Forming & Solving Equations**

Jan 2020 P2HR - Jan 2020 | P2HR

3 Three tins, A , B and C , each contain buttons.

Tin A contains x buttons.

Tin B contains 4 times the number of buttons that tin A contains.

Tin C contains 7 fewer buttons than tin A .

The total number of buttons in the three tins is 137

Work out the number of buttons in tin C .

(Total for Question 3 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Forming & Solving Equations**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Why it moved The question**

Why it moved: The question asks us to form and solve an equation for the numbers of buttons in three tins. There is no probability event.

2. Let the number of buttons

Let the number of buttons in tin A be x .

3. Solve equation

Then tin B has $4x$ buttons and tin C has $x - 7$ buttons.

$$x + 4x + (x - 7) = 137$$

$$6x - 7 = 137$$

$$6x = 144$$

$$x = 24$$

4. Tin C has

Tin C has

$$24 - 7 = 17$$

FINAL ANSWER

17 buttons.

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P2HR - Jan 2020 | P2HR

- 4 The diagram shows a rectangle and a diagonal of the rectangle.

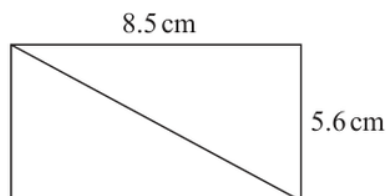


Diagram **NOT**
accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

..... cm

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 4** **Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$d^2 = 8.5^2 + 5.6^2$$

$$d = \sqrt{103.61} = 10.178 \dots$$

FINAL ANSWER

10.2 cm.

EA

PAST PAPER QUESTION**Q#: 5** **Marks: 3****TOPIC: Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

- 5** A plane takes 3 hours 36 minutes to fly from the Cayman Islands to New York.
The plane flies a distance of 2470 km.

Work out the average speed of the plane in km/h.
Give your answer correct to the nearest whole number.

..... km/h

(Total for Question 5 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The time is

The time is

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

2. Average speed

Average speed:

$$2470 \div 3.6 = 686.111 \dots$$

FINAL ANSWER

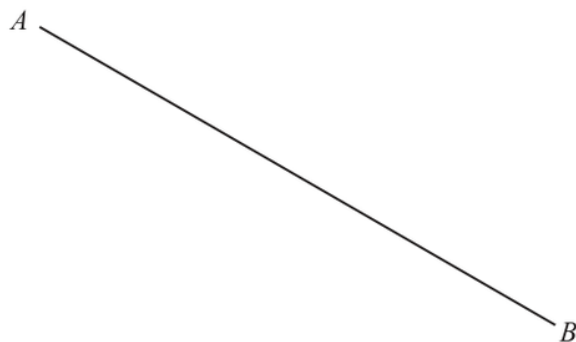
686 km/h.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

Jan 2020 P2HR - Jan 2020 | P2HR

- 6 Use ruler and compasses only to construct the perpendicular bisector of the line AB .
You must show all your construction lines.



(Total for Question 6 is 2 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the perpendicular bisector of AB :

1. Set the compass radius to more than half of AB .
2. With centre A , draw arcs above and below the line.
3. With the same radius and centre B , draw arcs crossing the first pair.
4. Draw the straight line through the two arc intersections.

FINAL ANSWER

This line is the perpendicular bisector of AB .

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Simultaneous Equations**

Jan 2020 P2HR - Jan 2020 | P2HR

7 Solve the simultaneous equations

$$\begin{aligned} 3x + 5y &= 6 \\ 7x - 5y &= -11 \end{aligned}$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 7** **Marks: 3****TOPIC: Simultaneous Equations**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$3x + 5y = 6$$

$$7x - 5y = -11$$

2. Add the equations

Add the equations:

$$10x = -5$$

$$x = -\frac{1}{2}$$

3. Solve simultaneous equationsSubstitute into $3x + 5y = 6$:

$$3\left(-\frac{1}{2}\right) + 5y = 6$$

$$-\frac{3}{2} + 5y = 6$$

$$5y = \frac{15}{2}$$

$$y = \frac{3}{2}$$

FINAL ANSWER

$$x = -\frac{1}{2}, y = \frac{3}{2}.$$

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2020 P2HR - Jan 2020 | P2HR

- 8** Hamish buys a new car for \$20 000
The car depreciates in value by 19% each year.
- Work out the value of the car at the end of 3 years.
Give your answer to the nearest \$.

\$.....

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. A depreciation of gives a**

A depreciation of 19% gives a multiplier of

$$1 - 0.19 = 0.81$$

2. Use compound interest

After 3 years,

$$20000(0.81)^3 = 10628.82$$

3. Use compound interest

Correct to the nearest dollar,

$$10628.82 \approx 10629$$

FINAL ANSWER**\$10629**

PAST PAPER QUESTION**Q#: 9** **Marks: 3****TOPIC: Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

- 9 The diagram shows a box in the shape of a cuboid.

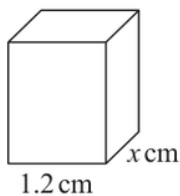


Diagram **NOT**
accurately drawn

The box is put on a table.

The face of the box in contact with the table has length 1.2 metres and width x metres.

The force exerted by the box on the table is 27 newtons.

The pressure on the table due to the box is 30 newtons/m²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the value of x .

$x = \dots\dots\dots$

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$30 = \frac{27}{\text{area}}$$

$$\text{area} = \frac{27}{30} = 0.9 \text{ m}^2$$

2. The base area is

The base area is

$$1.2x = 0.9$$

$$x = \frac{0.9}{1.2} = 0.75$$

FINAL ANSWER $x = 0.75 \text{ m.}$

PAST PAPER QUESTION**Q#: 10** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2020 P2HR - Jan 2020 | P2HR

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

- (a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean.
Give your answer in standard form.

..... square kilometres
(2)

The surface area of the Pacific Ocean is k times the surface area of the Arctic Ocean.

- (b) Work out the value of k .
Give your answer correct to the nearest whole number.

$k =$
(1)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Atlantic surface area**

Atlantic surface area:

$$1.06 \times 10^8 = 106000000$$

2. Indian surface area

Indian surface area:

$$6.86 \times 10^7 = 68600000$$

3. Difference

Difference:

$$106000000 - 68600000 = 37400000$$

$$37400000 = 3.74 \times 10^7$$

4. Convert standard form

For part (b),

$$k = \frac{1.56 \times 10^8}{1.41 \times 10^7}$$

$$k = 11.063 \dots$$

5. Convert standard form

To the nearest whole number,

$$k = 11$$

FINAL ANSWER(a) 3.74×10^7 , (b) 11

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Graphing Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

- 11** (a) Write down the integer values of x that satisfy the inequality $-2 < x \leq 4$

.....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

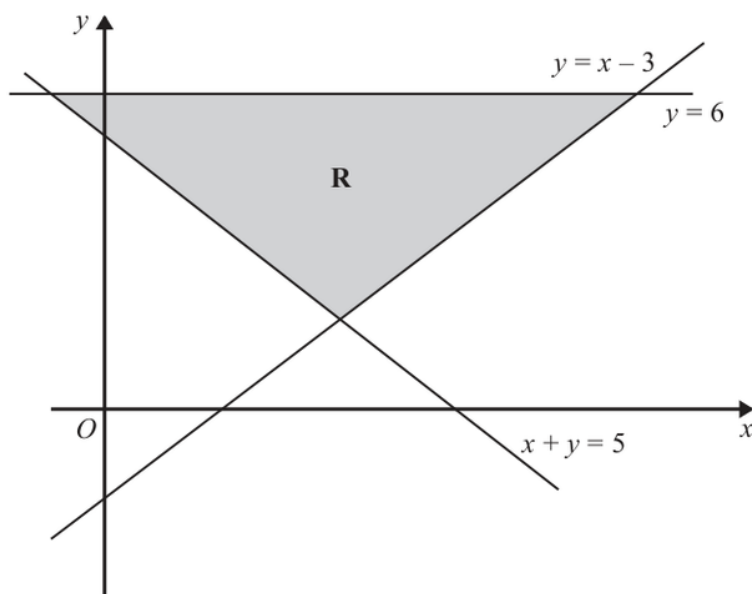


Diagram **NOT**
accurately drawn

- (b) Write down the three inequalities that define the region **R**.

.....
(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Graphing Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

For part (a),

$$2 < x \leq 4$$

2. The integer values are

The integer values are

$$3, 4$$

3. Calculate valueFor the region R , the boundaries are

$$y = x - 3, \quad x + y = 5, \quad y = 6$$

4. The shaded region is above

The shaded region is above the two sloping lines and below the horizontal line:

$$y \geq x - 3$$

$$x + y \geq 5$$

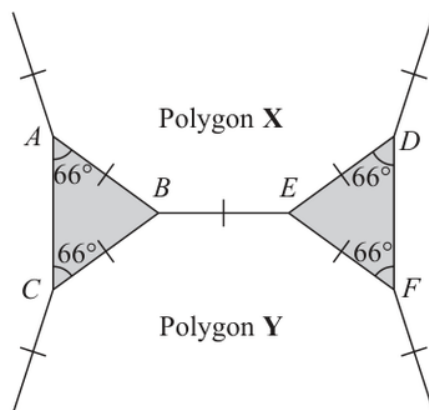
$$y \leq 6$$

FINAL ANSWER(a) 3, 4. (b) $y \geq x - 3$, $x + y \geq 5$, $y \leq 6$.

PAST PAPER QUESTION**Q#: 12** **Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Jan 2020 P2HR - Jan 2020 | P2HR

- 12** The diagram shows two congruent isosceles triangles and parts of two congruent regular polygons, **X** and **Y**.

Diagram **NOT**
accurately drawn

The two regular polygons each have n sides.

Work out the value of n .

$n = \dots\dots\dots$

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Each congruent isosceles triangle has**Each congruent isosceles triangle has two angles of 66° .**2. Use trigonometry**

So the angle at the point between the two polygons is

$$180^\circ - 66^\circ - 66^\circ = 48^\circ$$

3. The two regular polygons have

The two regular polygons have equal interior angles. Around the point:

$$48^\circ + I + I = 360^\circ$$

$$2I = 312^\circ$$

$$I = 156^\circ$$

4. The exterior angle is

The exterior angle is

$$180^\circ - 156^\circ = 24^\circ$$

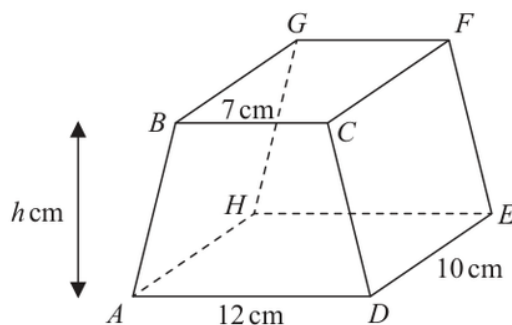
$$n = \frac{360}{24} = 15$$

FINAL ANSWER

$$n = 15.$$

PAST PAPER QUESTION**Q#: 13** **Marks: 3****TOPIC: Volume & Surface Area**

Jan 2020 P2HR - Jan 2020 | P2HR

13Diagram **NOT**
accurately drawn

The diagram shows a prism $ABCDEFGH$ in which $ABCD$ is a trapezium with BC parallel to AD and $CDEF$ is a rectangle.

$$BC = 7 \text{ cm} \quad AD = 12 \text{ cm} \quad DE = 10 \text{ cm}$$

The height of trapezium $ABCD$ is h cm

The volume of the prism is 608 cm^3

Work out the value of h .

$$h = \dots\dots\dots$$

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Volume & Surface Area**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Area of the trapezium cross

Area of the trapezium cross-section:

$$\frac{1}{2}(7 + 12)h = 9.5h$$

2. Volume of the prism

Volume of the prism:

$$9.5h \times 10 = 608$$

$$95h = 608$$

$$h = 6.4$$

FINAL ANSWER

$$h = 6.4 \text{ cm.}$$

PAST PAPER QUESTION**Q#: 14** **Marks: 3****TOPIC: Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term.

Here are his marks.

18 5 7 12 11 18 15 16 17 13 14

Find the interquartile range of the marks.

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Put the marks in order**

Put the marks in order:

5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18

$$Q_1 = 11, \quad Q_3 = 17$$

$$\text{IQR} = 17 - 11 = 6$$

FINAL ANSWER

6.

EA

PAST PAPER QUESTION**Q#: 15** **Marks: 4****TOPIC: Graphs of Functions**

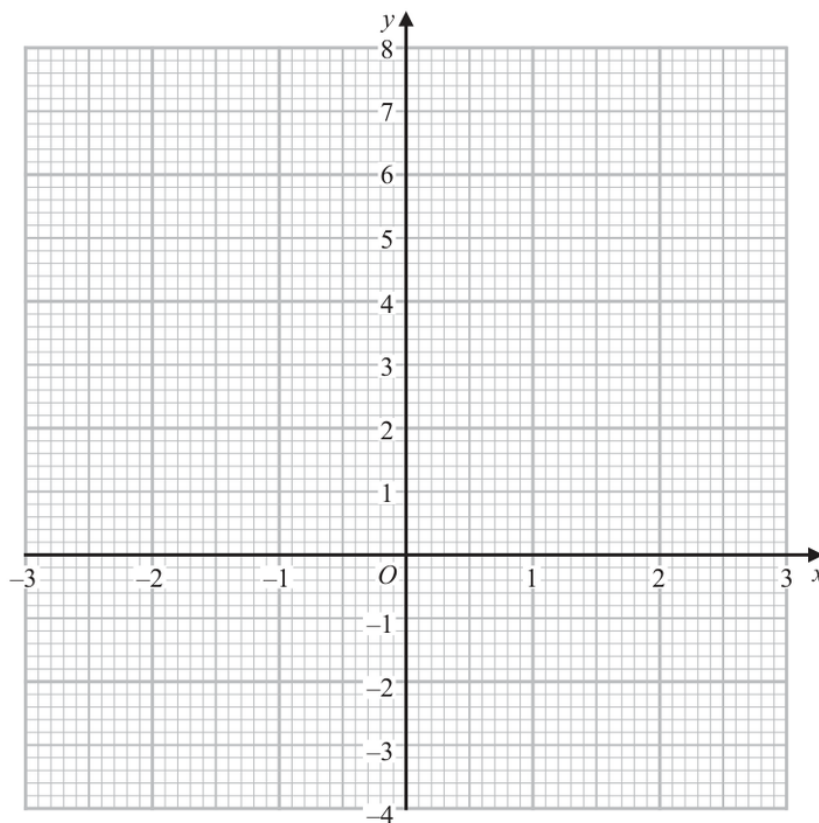
Jan 2020 P2HR - Jan 2020 | P2HR

15 (a) Complete the table of values for $y = x^2 - \frac{x}{2} - 3$

x	-3	-2	-1	0	1	2	3
y	7.5				-2.5		4.5

(2)

(b) On the grid, draw the graph of $y = x^2 - \frac{x}{2} - 3$ for values of x from -3 to 3



(2)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15** **Marks: 4****TOPIC: Graphs of Functions**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve quadratic equation**

$$y = x^2 - \frac{x}{2} - 3$$

2. The completed table is

The completed table is:

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

3. Plot these points and join

Plot these points and join them with a smooth U-shaped curve.

FINAL ANSWER

Missing values are 2, -1.5, -3, 0.

PAST PAPER QUESTION**Q#: 16****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

16 Cody has two bags of counters, bag A and bag B.

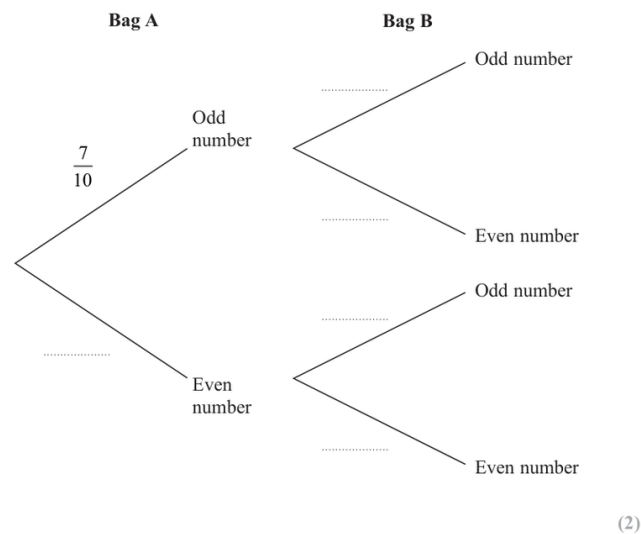
Each of the counters has either an odd number or an even number written on it.

There are 10 counters in bag A and 7 of these counters have an **odd** number written on them.

There are 12 counters in bag B and 7 of these counters have an **odd** number written on them.

Cody is going to take at random a counter from bag A and a counter from bag B.

(a) Complete the probability tree diagram.



(b) Calculate the probability that the total of the numbers on the two counters will be an odd number.

(3)

Harriet also has a bag of counters.

Each of her counters also has either an odd number or an even number written on it.

Harriet is going to take at random a counter from her bag of counters.

The probability that the number on each of Cody's two counters **and** the number on

Harriet's counter will all be even is $\frac{3}{100}$

(c) Find the least number of counters that Harriet has in her bag.
Show your working clearly.

(3)

(Total for Question 16 is 8 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

For Cody:

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

$$P(B \text{ odd}) = \frac{7}{12}, \quad P(B \text{ even}) = \frac{5}{12}$$

2. Calculate probability

An odd total comes from odd + even or even + odd:

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

3. Cody takes two even counters

Cody takes two even counters with probability

$$\frac{3}{10} \times \frac{5}{12} = \frac{1}{8}$$

4. Given Cody and Harriet both

Given Cody and Harriet both take two even counters:

$$\frac{1}{8} \times P(H \text{ even}) = \frac{3}{100}$$

$$P(H \text{ even}) = \frac{6}{25}$$

5. Calculate probability

So Harriet's bag must have at least 25 counters.

FINAL ANSWER $\frac{7}{15}$, and 25.

PAST PAPER QUESTION**Q#: 17** **Marks: 6****TOPIC: Set Notation & Venn Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

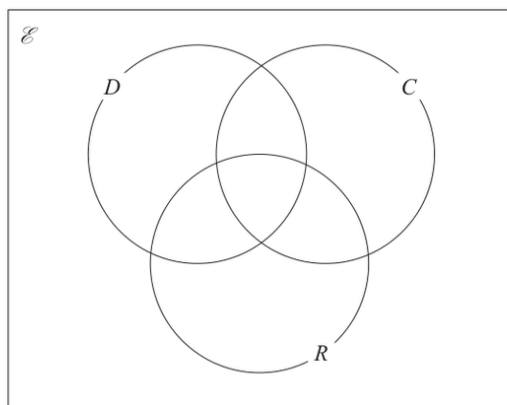
17 Some students in a school were asked the following question.

“Do you have a dog (D), a cat (C) or a rabbit (R)?”

Of these students

- 28 have a dog
- 18 have a cat
- 20 have a rabbit
- 8 have both a cat and a rabbit
- 9 have both a dog and a rabbit
- x have both a dog and a cat
- 6 have a dog, a cat and a rabbit
- 5 have not got a dog or a cat or a rabbit

- (a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of x where necessary.



(3)

Given that a total of 50 students answered the question,

- (b) work out the value of x .

$x =$
(2)

- (c) Find $n(C' \cap D')$

.....
(1)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 6****TOPIC: Set Notation & Venn Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the sets dogs**Let the sets be D for dogs, C for cats, and R for rabbits.**2. The number in all three**

The number in all three sets is 6.

$$D \cap R \text{ only} = 9 - 6 = 3$$

$$C \cap R \text{ only} = 8 - 6 = 2$$

$$D \cap C \text{ only} = x - 6$$

3. Calculate value

So the remaining one-set regions are

$$D \text{ only} = 25 - x, \quad C \text{ only} = 16 - x, \quad R \text{ only} = 9$$

4. There are students outside all

There are 5 students outside all three sets. The total number of students is 50, so

$$(25 - x) + (16 - x) + 9 + (x - 6) + 3 + 2 + 6 + 5 = 50$$

$$60 - x = 50$$

$$x = 10$$

5. Calculate valueFor $C' \cap D'$, we need students who are not in C and not in D . These are the rabbit-only students and the students outside all three sets.

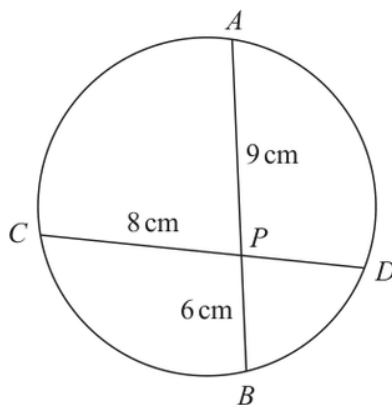
$$9 + 5 = 14$$

FINAL ANSWER

$$x = 10 \text{ and } n(C' \cap D') = 14$$

PAST PAPER QUESTION**Q#: 18****Marks: 2****TOPIC: Circle Theorems**

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18Diagram **NOT**
accurately drawn APB and CPD are chords of a circle. $AP = 9\text{ cm}$ $PB = 6\text{ cm}$ $CP = 8\text{ cm}$ Calculate the length of PD .

..... cm

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 2****TOPIC: Circle Theorems**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use the intersecting chords theorem**

Use the intersecting chords theorem:

$$AP \times PB = CP \times PD$$

2. Substitute the values

Substitute the values:

$$9 \times 6 = 8 \times PD$$

$$54 = 8PD$$

$$PD = 6.75$$

FINAL ANSWER

6.75 cm.

PAST PAPER QUESTION**Q#: 19** **Marks: 6****TOPIC: Solving Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

19 (a) Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

(b) Solve the inequality $5y^2 - 17y \leq 40$

$$\dots\dots\dots (3)$$

(Total for Question 19 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 6****TOPIC: Solving Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\frac{4 - 3x}{5} - \frac{3x - 5}{2} = -3$$

2. Multiply by 10

Multiply by 10:

$$2(4 - 3x) - 5(3x - 5) = -30$$

$$8 - 6x - 15x + 25 = -30$$

$$33 - 21x = -30$$

$$-21x = -63$$

$$x = 3$$

3. (b)

(b)

$$5y^2 - 17y \leq 40$$

$$5y^2 - 17y - 40 \leq 0$$

4. Factorise

Factorise:

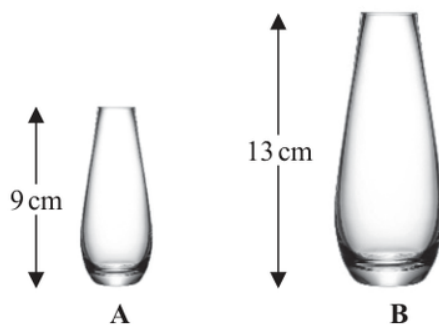
$$(5y + 8)(y - 5) \leq 0$$

5. Solve quadratic equationThe roots are $-\frac{8}{5}$ and 5. The quadratic is ≤ 0 between the roots.**FINAL ANSWER**

$$x = 3, \text{ and } -\frac{8}{5} \leq y \leq 5.$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

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20 The diagram shows two similar vases, **A** and **B**.The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**...... cm²

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The two vases are similar**

The two vases are similar.

2. Their heights are in the

Their heights are in the ratio

$$9 : 13$$

3. Split the ratio

So their surface areas are in the ratio

$$9^2 : 13^2 = 81 : 169$$

4. The total surface area isThe total surface area is 1800 cm^2 , so there are

$$81 + 169 = 250$$

5. parts altogether

parts altogether.

6. Surface area of vase A

Surface area of vase A:

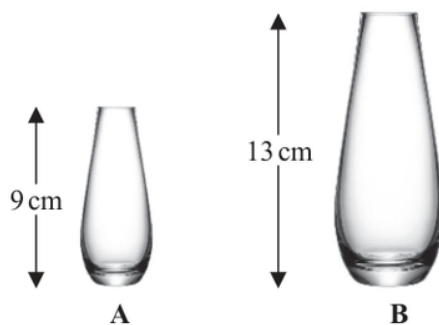
$$1800 \times \frac{81}{250} = 583.2$$

FINAL ANSWER

$$583.2 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

20 The diagram shows two similar vases, **A** and **B**.Diagram **NOT**
accurately drawnThe height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**...... cm²

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

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6. Surface area of vase A

Surface area of vase A:

$$1800 \times \frac{81}{250} = 583.2$$

FINAL ANSWER

$$583.2 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 21****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

21 (a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

.....
(3)

(b) Find an expression for n in terms of y .
Show clear algebraic working and simplify your expression.

.....
(4)

(Total for Question 21 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factorise expression**

For part (a), factorise the numerator and denominator:

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

$$4x^2 - 9 = (2x - 3)(2x + 3)$$

2. Use index laws

So

$$\begin{aligned} \frac{10x^2 + 23x + 12}{4x^2 - 9} &= \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)} \\ &= \frac{5x + 4}{2x - 3} \end{aligned}$$

3. Use index laws

For part (b),

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

4. Use index laws

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

5. Equate powers of 2

Equate powers of 2:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

FINAL ANSWER(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

PAST PAPER QUESTION**Q#: 21****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

21 (a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

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(b) Find an expression for n in terms of y .
Show clear algebraic working and simplify your expression.

.....
(4)

(Total for Question 21 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

$$= \frac{5x + 4}{2x - 3}$$

2. (b)

(b)

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

3. Evaluate fraction

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

4. Equate powers

Equate powers:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

FINAL ANSWER

$$(a) \frac{5x + 4}{2x - 3}, (b) n = 5y - 1$$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

- 22** The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865

Find the 9th term of S .

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the nth term**

First term:

$$a = -6$$

$$S_{30} = 2865$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2865 = \frac{30}{2}(2(-6) + 29d)$$

$$2865 = 15(-12 + 29d)$$

$$191 = -12 + 29d$$

$$29d = 203$$

$$d = 7$$

3. The 9th term is

The 9th term is

$$u_9 = a + 8d$$

$$u_9 = -6 + 8(7) = 50$$

FINAL ANSWER

50

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

- 22** The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865

Find the 9th term of S .

(Total for Question 22 is 4 marks)

EA

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Jan 2020 P2HR - Jan 2020 | P2HR

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The 9th term is

$$u_9 = a + 8d$$

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FINAL ANSWER

50

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 23** **Marks: 3****TOPIC: Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

$$= -2(x^2 + 6x) + 7$$

$$= -2((x + 3)^2 - 9) + 7$$

$$= -2(x + 3)^2 + 18 + 7$$

$$= 25 - 2(x + 3)^2$$

FINAL ANSWER

$$25 - 2(x + 3)^2$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 23** **Marks: 3****TOPIC: Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

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$$= 25 - 2(x + 3)^2$$

FINAL ANSWER

$$25 - 2(x + 3)^2$$

PAST PAPER QUESTION**Q#: 24** **Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

24 L_1 and L_2 are two straight lines.

The origin of the coordinate axes is O .

L_1 has equation $5x + 10y = 8$

L_2 is perpendicular to L_1 and passes through the point with coordinates $(8, 6)$

L_2 crosses the x -axis at the point A .

L_2 intersects the straight line with equation $x = -3$ at the point B .

Find the area of triangle AOB .

Show your working clearly.

(Total for Question 24 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**For L_1 ,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

2. Find the gradientSo the gradient of L_1 is $-\frac{1}{2}$.**3. Find the gradient**Since L_2 is perpendicular to L_1 , the gradient of L_2 is 2.**4. Calculate value**

Using the point (8, 6):

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

5. Calculate valueAt A, the line crosses the x -axis, so $y = 0$:

$$0 = 2x - 10$$

$$x = 5$$

6. Calculate valueSo $A = (5, 0)$.**7. Calculate value**At B, $x = -3$:

$$y = 2(-3) - 10 = -16$$

8. Calculate valueSo $B = (-3, -16)$.**9. Triangle has base and perpendicular**

Triangle AOB has base $OA = 5$ and perpendicular height 16.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

FINAL ANSWER

40

EA

PAST PAPER QUESTION

Q#: 24

Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

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Jan 2020 P2HR - Jan 2020 | P2HR

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Triangle AOB has base $OA = 5$ and perpendicular height 16.

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FINAL ANSWER

40

EA

PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

25 N is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**Since N is a multiple of 5, let

$$N = 5k$$

2. where is an integerwhere k is an integer.**3. Given**

Given

$$A = N + 1, \quad B = N - 1$$

4. Use trigonometry

So

$$A^2 - B^2 = (N + 1)^2 - (N - 1)^2$$

5. Expand

Expand:

$$(N + 1)^2 = N^2 + 2N + 1$$

$$(N - 1)^2 = N^2 - 2N + 1$$

6. Use trigonometry

Therefore

$$\begin{aligned} A^2 - B^2 &= (N^2 + 2N + 1) - (N^2 - 2N + 1) \\ &= 4N \end{aligned}$$

7. SubstituteSubstitute $N = 5k$:

$$4N = 4(5k) = 20k$$

8. Use trigonometrySince k is an integer, $20k$ is a multiple of 20.

FINAL ANSWER

$A^2 - B^2$ is always a multiple of 20.

EA

PAST PAPER QUESTION

Q#: 25

Marks: 3

TOPIC: **Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

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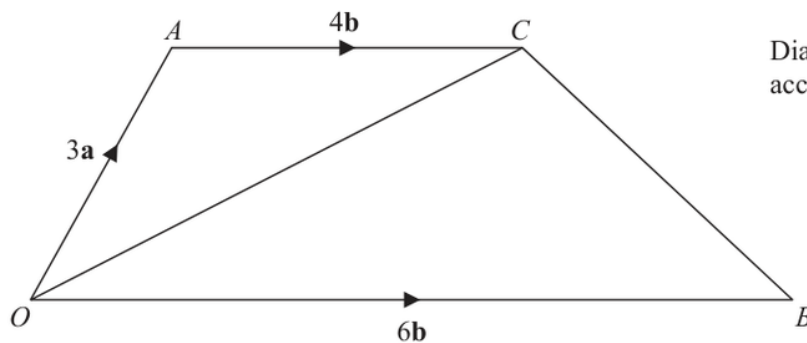
FINAL ANSWER

$A^2 - B^2$ is always a multiple of 20.

EA

PAST PAPER QUESTION**Q#: 26****Marks: 5**TOPIC: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

26 The diagram shows trapezium $OACB$.Diagram **NOT**
accurately drawn

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AC} = 4\mathbf{b}$$

 N is the point on OC such that ANB is a straight line.Find $\vec{ON}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

2. Read the graph

So

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

3. Read the graphPoint N lies on OC , so

$$\vec{ON} = t(3\mathbf{a} + 4\mathbf{b})$$

4. Read the graphPoint N also lies on AB , so

$$\vec{ON} = 3\mathbf{a} + s(6\mathbf{b} - 3\mathbf{a})$$

$$\vec{ON} = (3 - 3s)\mathbf{a} + 6s\mathbf{b}$$

5. Equate coefficients

Equate coefficients:

$$3t = 3 - 3s$$

$$4t = 6s$$

6. Read the graph

So

$$t = 1 - s$$

7. Read the graph

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

8. Read the graph

Then

$$t = \frac{3}{5}$$

9. Read the graph

Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

FINAL ANSWER

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

EA

PAST PAPER QUESTION

Q#: 26

Marks: 5

TOPIC: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

26 The diagram shows trapezium $OACB$.

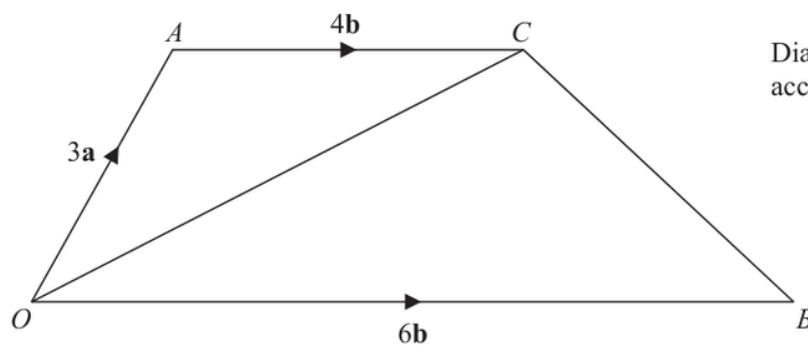


Diagram **NOT**
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N is the point on OC such that ANB is a straight line.

Find $\vec{ON}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$.

(Total for Question 26 is 5 marks)

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Jan 2020 P2HR - Jan 2020 | P2HR

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Then

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Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

FINAL ANSWER

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$